# Gen Li

*Final-year PhD candidate at University of Edinburgh*

li.gen@ed.ac.uk | Homepage | Github | Google Scholar

## EDUCATION

**University of Edinburgh** *2021.09 - 2025.07 (Expected)*
*Ph.D. Robotics and Autonomous Systems - Edinburgh, UK*

**Sungkyunkwan University** *2018.09 - 2020.08*
*M.S. Electrical and Computer Engineering - Suwon, South Korea*

**Xidian University** *2014.08 - 2018.06*
*B.S. Electronic and Information Engineering - Xi'an, China*

## RESEARCH FIELDS

The central focus of my research is to empower machines to perceive and interact with their environments in a data-efficient manner. This includes exploring a range of cutting-edge topics:

- **Embodied AI -** Affordance Learning, Robotic Manipulation
- **Computer Vision -** Scene Understanding, Vision-Language Models
- **Efficient Learning -** Learning under Weak Supervision & Limited Data
- **Human-Object Interaction** - Egocentric Vision, Interaction Understanding / Generation

## RESEARCH EXPERIENCE

**Affordance Learning & Human-Object Interaction Understanding** *2021.09 - Present*
*PhD's Research, Supervisor: Prof. Laura Sevilla-Lara*

**Prototypical Learning for Few-Shot Semantic Segmentation** *2020.10 - 2021.08*
*Visiting Researcher at the UoE, Supervisor: Prof. Laura Sevilla-Lara*

**Lightweight Real-time Semantic Segmentation Network** *2019.06 - 2020.07*
*Master's Research, Supervisor: Prof. Joongkyu Kim*

**Deep Learning based Semantic Segmentation** *2017.11 - 2018.06*
*Undergraduate Thesis, Supervisor: Prof. Cheolkon Jung*

## SELECTED PUBLICATIONS (total citations: 950+)

- One-Shot Open Affordance Learning with Foundation Models [project]
  **Gen Li**, Deqing Sun, Laura Sevilla-Lara, Varun Jampani
  *CVPR 2024*
- LOCATE: Localize and Transfer Object Parts for Weakly Supervised Affordance Grounding
  **Gen Li**, Varun Jampani, Deqing Sun, Laura Sevilla-Lara [project]
  *CVPR 2023*
- Adaptive Prototype Learning and Allocation for Few-Shot Segmentation [project]
  **Gen Li**, Varun Jampani, Laura Sevilla-Lara, Deqing Sun, Jonghyun Kim, Joongkyu Kim
  *CVPR 2021* ***[410+ citations]***
- DABNet: Depth-wise Asymmetric Bottleneck for Real-time Semantic Segmentation [paper] [code]
  **Gen Li**, Joongkyu Kim
  *BMVC 2019* ***[310+ citations]***
- Depth-wise Asymmetric Bottleneck with Point-wise Attention Decoder for Real-time Semantic Segmentation in Urban Scenes [paper]
  **Gen Li**, Shenlu Jiang, Inyong Yun, Jonghyun Kim, Joongkyu Kim
  *IEEE Access (SCI, IF: 4.098)*
- Weakly Supervised Temporal Attention 3D Network for Human Action Recognition [paper]
  Jonghyun Kim, **Gen Li**, Inyong Yun, Cheolkon Jung, Joongkyu Kim
  *Pattern Recognition (SCI, IF: 7.196)*

- Edge and Identity Preserving Network for Face Super-Resolution [paper] [code]
  Jonghyun Kim, **Gen Li**, Inyong Yun, Cheolkon Jung, Joongkyu Kim
  *Neurocomputing (SCI, IF: 4.438)*
- Referenceless User Controllable Semantic Image Synthesis [paper]
  Jonghyun Kim, **Gen Li**, Joongkyu Kim
  *IJCNN 2023*
- SuperStyleNet: Deep Image Synthesis with Superpixel Based Style Encoder [paper] [code]
  Jonghyun Kim, **Gen Li**, Cheolkon Jung, Joongkyu Kim
  *BMVC 2021*

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### Preprint

- Learning Precise Affordances from Egocentric Videos for Robotic Manipulation [project]
  **Gen Li**, Nikolaos Tsagkas, …, Sethu Vijayakumar, Kun Shao, Laura Sevilla-Lara
- Robotic Embodied Intelligence: Integrating Language Models, Curated Knowledge, and Multimodal Feedback for Complex Task Success [paper]
  Ruaridh Mon-Williams, **Gen Li**, Ran Long, Wenqian Du, Chris Lucas
- Principles of Visual Tokens for Efficient Video Understanding [paper]
  Xinyue Hao, **Gen Li**, Shreyank N Gowda, …, Anurag Arnab, Laura Sevilla-Lara

## WORK EXPERIENCE

**Huawei, Noah's Ark Lab, London** *2023.07 - 2024.02*
Research Intern | Affordance-Oriented Robotic Manipulation

## ACADEMIC SERVICE

**Reviewer**

- Conference - CVPR2024/2025, AAAI2024, ECCV 2024
- Journal - IEEE Transactions on Image Processing, Neurocomputing

**Teaching**

- [2023 - 2024] Introductory Applied Machine Learning - Demonstrator & Tutor
- [2022 - 2024] Applied Machine Learning - Teaching Assistant
- [2018 - 2020] Digital Signal Processing - Teaching Assistant

## AWARDS AND GRANTS

- Ph.D. - EPSRC funded CDT-RAS Scholarship, 100% tuition waiver and an annual stipend
- M.S. - STEM Scholarship, 100% tuition waiver, Samsung Funding
- State Scholarship for Study Abroad, China Scholarship Council of P.R. China

## PROFESSIONAL SKILLS

- Programming Language: Python, C/C++, LaTeX, Matlab, HTML/CSS, etc.
- Frameworks and Tools: Pytorch/Tensorflow/Keras, Git, Docker, Linux Bash, etc.